

JINKI JUNG

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GitHub: <https://github.com/Jinkijung>
citations : <http://scholar.google.co.kr/citations?user=inzigzUAAAAJ&hl=en&authuser=1>

HIGHLIGHT

Jinki Jung is a senior software developer of Digital Maritime Consultancy, contributing mostly technical aspects of Maritime Connectivity Platform (MCP), one of the biggest open-source based platform in maritime industry, as a system architect and a full-stack developer. Previously he was a researcher and developer of Augmented Reality/Virtual Reality/Human-Computer interaction area addressing interaction and architecture for the use of those in mobile environment. In his free time, he contributes an open-source project named iil with his interest in modeling all kinds of work and automation powered by iil.

CAREER

Digital Maritime Consultancy, Denmark

Senior Software Developer, Secretariat of MCP

Support each Topic: Massive Ship Traffic Simulation

2019-Now

Korea Research Institute of Ships and Ocean Engineering (KRISO), Republic of Korea

Postdoctoral researcher, Maritime Safety Research Division

Lead researcher of Virtual Reality / Augmented Reality for Safety use (e.g., maritime training)

2016-2019

Korea Advanced Institute of Science and Technology (KAIST), Republic of Korea

Postdoctoral researcher, Information and Electronics Research Institute

Lead researcher for addressing user-centric interaction (e.g., hand gestures)

and performance optimization in Augmented reality

2015-2016

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D., Computer Science

Research Topic: Mobile Augmented Reality

Dissertation: "Real-time Sensor Fusion based Mobile Augmented Reality Framework"

Advisor: Hyun S. Yang

2009-2015

Korea Advanced Institute of Science and Technology (KAIST)

M.S., Computer Science 2009

Research Topic: Natural User Interface for Augmented Reality

Thesis: "A Real-time Robust Body Part Tracking System for Intelligent Environment"

Advisor: Hyun S. Yang

2007-2009

Soongsil University

B.S., Media Engineering 2007

2003-2007

PROJECTS

Maritime Connectivity Platform (MCP) - <https://maritimeconnectivity.net>

System architect / technical advisor / code contributor / WG member (<https://github.com/maritimeconnectivity>) 2018 – Now

Contribute to develop the architecture of distributed microservices of MCP core components and implement of the components including Maritime Identity Registry (PKI, digital certificate, OIDC, HSM and MIR specification/installation/documentation), Maritime Service Registry (Distributed ledger / Blockchain, Spring boot application backend and React-based web front-end) and Maritime Messaging Service (high-level description / specification for standardization)

Maritime Resource Registry (MRR) - <https://digital-maritime-consultancy.github.io/mrr-portal/>

System architect / front-end main contributor (<https://github.com/Digital-Maritime-Consultancy/mrr-portal>)

2021 – Now

Contribute to build the architecture of MRR for storing various maritime resources annotated with Maritime Resource Name (IALA R1023) and version and looking up with the information as such.

Currently MRR is a prototype to introduce the concept to relevant organizations and stakeholders and will be deployed as an operational service.

VoTT-dot: point-based annotation support for VoTT - <https://github.com/Digital-Maritime-Consultancy/VoTT-dot>

Lead developer for both front- and back-end parts

2021 – Now

Update the VoTT annotation tool of Microsoft for supporting point-based annotation and other advanced features, described in detail at the VoTT-dot readme file.

Public open big data: Maritime Object AI Data - <https://aihub.or.kr/aidata/34155>

Annotation tool & DL developer (<https://github.com/Digital-Maritime-Consultancy/VoTT/tree/0.7>)

2020 – Dec. 2020

Develop an annotation tool for Korean national project establishing a maritime object image database and a panoptic segmentation model for validating the database

iiI: a work description - <https://jinkijung.github.io/iiI-docs/>

Founder / main contributor

2020 – Now

Propose and develop iiI, a work description model, for managing both personal and collaborative work efficiently.

TECHNICAL SKILLS

Programming Language: C, C++, Java, Typescript, Objective C, Python, Javascript

Front-end framework: React, Angular

Back-end framework: Spring boot, Node.js, NestJS

Graphics Engine: Unity3D, UE5, OpenGL, WebGL

Mobile Programming: iOS, Android

PUBLICATIONS

International Journals

Discipline vs guidance: comparison of visual engagement approaches in immersive virtual environments

Hyeopwoo Lee, **Jinki Jung**, Heung Kyu Lee, Hyun Seung Yang

Multimedia tools and applications, 2021

Effects of interface on procedural skill transfer in virtual training: Lifeboat launching operation study

Jinki Jung, Young Joong Ahn

Computer Animation and Virtual Worlds, 2018

Efficient mobile AR technology using scalable recognition and tracking based on server-client model

Jinki Jung, Jaewon Ha, Sang-Wook Lee, Francisco A Rojas, Hyun S Yang

Computers & Graphics, 2012

Real-time recognition and tracking for augmented reality books

Kyusung Cho, **Jinki Jung**, Sang-Wook Lee, Sang Ok Lim, Hyun Seung Yang

Computer Animation and Virtual Worlds, 2011

Domestic Journal

A Study on the Implementation of a Web-browser-based Global e-Navigation Service Discovery System for Decentralized Maritime Service Registries

Jinki Jung, Young Joong Ahn

The Journal of Navigation and Port Research, 2022

A Virtual Sailor Training Platform for Fire Drills on Ship

Jinki Jung, Jin Hyung Park

The Journal of Navigation and Port Research, 2016

Real-time Sensor-aided Scene Analysis based on Line Graph for Mobile Augmented Reality

Jinki Jung

The Journal of Korean Institute of Information Technology, 2015

International Conferences

Annotation vs. Virtual Tutor: Comparative Analysis on the Effectiveness of Visual Instructions in Immersive Virtual Reality

Hyeopwoo Lee, Hyejin Kim, Diego Vilela Monteiro, Youngnoh Goh, Daseong Han, Hai-Ning Liang, Hyun Seung Yang, and **Jinki Jung**

18th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2019)

Ensuring Safety in Augmented Reality from Trade-off Between Immersion and Situation Awareness

Jinki Jung, Hyeopwoo Lee, Jeehye Choi, Abhilasha Nanda, Uwe Grünefeld, Tim Claudius Stratmann, Wilko Heuten

17th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2018)

Guiding Smombies: Augmenting Peripheral Vision with Low-Cost Glasses to Shift the Attention of Smartphone Users

Uwe Grünefeld, Tim Claudius Stratmann, **Jinki Jung**, Hyeopwoo Lee, Jeehye Choi, Abhilasha Nanda, Wilko Heuten

17th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2018)

An Adaptive Augmented Reality Interface for Hand based on Probabilistic Approach

Jinki Jung, Hyeopwoo Lee, and Hyun Seung Yang

14th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2015)

Real-time sensor-fusion based Indoor Localization for Mobile Augmented Reality

Jinki Jung, Hyeopwoo Lee, Luis Weruaga, Jamal Zemerly and Hyun Seung Yang

20th International Conf. on Virtual Systems and Multimedia (VSMM 2014)

Augmented Keyboard: a Virtual Keyboard Interface for Smart glasses

Jinki Jung, Jinwoo Jeon, Hyeopwoo Lee, Kichan Kwon, Jamal Zemerly, Hyun S Yang

ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry (VRCAI 2014)

Smartphone as an augmented reality authoring tool via multi-touch based 3D interaction method

Jinki Jung, Jihye Hong, Sungheon Park, Hyun S Yang

ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry (VRCAI 2012)

Efficient 3D content authoring framework based on mobile AR

Sang-Wook Lee, **Jinki Jung**, Jihye Hong, Suwon Lee, Hyunwoo Cho, Hyun Seung Yang

18th International Conf. on Virtual Systems and Multimedia (VSMM 2012)

AR paint: a fusion system of a paint tool and AR

Suwon Lee, **Jinki Jung**, Jihye Hong, JB Ryu, Hyun S Yang

International Conference on Entertainment Computing (ICEC 2012)

AR postcard: the augmented reality system with a postcard

Hyunwoo Cho, **Jinki Jung**, Kyusung Cho, Yong-Ho Seo, Hyun S Yang

ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry (VRCAI 2011)

Mobile Augmented Reality using scalable recognition and tracking

Jaewon Ha, **Jinki Jung**, ByungOk Han, Kyusung Cho, Hyun Seung Yang

Virtual Reality Conference (VR 2011), IEEE

Online scene modeling for interactive AR applications

Jaesang Yoo, Kyusung Cho, **Jinki Jung**, Hyun S Yang

International Conference on Entertainment Computing (ICEC 2010)

Multiple page recognition and tracking for augmented books

Kyusung Cho, Jaesang Yoo, **Jinki Jung**, Hyun S Yang

International Conference on Entertainment Computing (ICEC 2010)

Real-time robust body part tracking for augmented reality interface

Jinki Jung, Kyusung Cho, Hyun S Yang

ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry (VRCAI 2009)

Hybrid visual tracking for augmented books

Hyun S Yang, Kyusung Cho, Jaemin Soh, **Jinki Jung**, Junseok Lee
International Conference on Entertainment Computing (ICEC 2009)

PATENTS

A virtual keyboard based on hand recognition and implementing method thereof
Korea Patent, 10-1559424, Co-inventor, 2015

3D interaction method for Augmented Reality using multi-touch interface
Korea Patent, 10-1338958, Co-inventor, 2013

Augmented reality system and method of a printed matter and video
Korea Patent, 10-1197126, Co-inventor, 2012

Efficient 3D object recognition using a tree structure
Korea Patent, 10-1068465, Co-inventor, 2011

3D OBJECT RECOGNITION SYSTEM AND METHOD

US Patent Pending, 12/912,211, Co-inventor, 2010

RESEARCH EXPERIENCES

Virtuality for Safety

Lead researcher

March 2018 – February 2019

Conducted researches on technology-driven safety ensuring components based on VR/AR

VR-simulated Sailor Training Platform for Emergency

Project manager, Researcher

June 2015 – February 2019

Developed a sailor training platform for emergency based on VR system that provides natural interaction with virtual ship environment.

KAIST-KUSTAR international joint research of indoor localization and context-aware Augmented Reality

Project manager, Researcher

March 2013 – June 2015

Designed real-time indoor localization using indoor scene analysis that makes use of line feature and multiple sensors of a mobile device.

Augmented Keyboard: a virtual keyboard interface for smart glasses

Project manager, Researcher

April 2014 – December 2014

Proposed a novel interaction design and developed a model-based hand palm tracking method for the proposed interface.

Performance optimization of markerless tracking module for mobile devices

Project manager, System engineer

March 2014 – December 2014

Doubled the tracking performance (FPS) on a mobile device by adapting multi-threaded architecture.

Probabilistic modeling of line-pair appearance for drawing image retrieval

Project manager, Researcher

September 2012 – March 2013

Proposed a novel histogram based matching method that employs the invariant properties of the pairwise line features.

Real-time 6DOF hand pose estimation for Augmented Reality using RGB camera

Researcher

December 2010 – June 2011

Developed the appearance based hand pose estimation by modeling spatial structure of skin colored region.

Real-time robust markerless recognition and tracking for Augmented Reality book

Researcher

March 2008 – January 2011

Developed the robust recognition performance against cluttered background by dividing the screen into multiple areas and separating the corresponding recognition results

Real-time body part tracking and gesture recognition for an intelligent environment

Project manager, Researcher

Proposed the RGB-image based 3D body part tracking method by using Kalman filter.

August 2008 – September 2009

PhotoGeo: 3D reconstruction of vehicles based on image based rendering modeling (IBMR)

System engineer

Implemented the texturing module and GUI interface of the system.

March 2007 – November 2007

INVITED TALKS

International Workshop on Intelligent Software Engineering 2022

Title: Sharing open source software development experiences

LANGUAGES

English – speak fluently and read/write with high proficiency

Korean – native language

Danish – beginner level
